

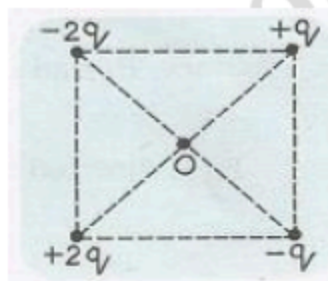
NUMERICAL PROBLEMS IN ELECTRIC CHARGES & FIELDS

MANDATORY PROBLEMS

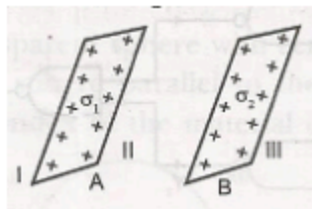
1. A proton and an electron are placed 1.6 cm apart in free space. Find the magnitude of electrostatic force between them. What is the nature of this force?
2. Two identical metal spheres, having unequal, opposite charges are placed at a distance 0.90 m apart in air. After bringing them in contact with each other, they are again placed at the same distance apart. Now the force of repulsion between them is 0.025 N. Calculate the final charge on each of them.
3. Calculate the Coulomb force between two α - particles separated by a distance of 3.2×10^{-15} m in air.
4. (a) Two insulated charged copper spheres A and B have their centres separated by a distance of 50 cm. What is the mutual force of electrostatic repulsion if the charge on each is 6.5×10^{-7} C? The radii of A and B are negligible compared to the distance of separation.

(b) What is the force of repulsion if
(i) each sphere is charged double the above amount, and the distance between them is halved.
(ii) the two spheres are placed in water? (Dielectric constant of water = 80)
5. A pendulum of mass 80 mg and carrying a charge of 20 nC is at rest in a horizontal Uniform electric field of 2×10^4 N/C. Find the tension in the thread of the pendulum and the angle it makes with the vertical.
6. Two point charges of $+5 \times 10^{-19}$ C and $+20 \times 10^{-19}$ C are separated by a distance of 2m. Find the point on the line joining them at which electric field intensity is zero.
7. Two point charges, each of 5 μ C but opposite in sign, are placed 4 cm. apart. Calculate the electric field intensity at a point distant 4cm from the midpoint on the axial line of the dipole.
8. An electric dipole of dipole moment 4×10^{-5} C m, $E = 10^3$ N/C, making an angle of 30° with the direction of the field. Determine the torque exerted by the electric field on the dipole.
9. An electric dipole of length 4 cm, when placed with its axis making an angle of 60° with a uniform electric field, experiences a torque of $4\sqrt{3}$ N m. Calculate: (i) magnitude of the electric field (ii) potential energy of the dipole if the dipole has charges of ± 8 nC.
10. An infinite line charge produces a field of 9×10^4 N/C at a distance of 4 cm. Calculate the linear charge density.
11. Maximum torque acting on an electric dipole of moment 3×10^{-29} Cm in a uniform electric field E is 6×10^{-25} Nm. Find E.

12. Intensity of electric field at a perpendicular distance of 0.5 m from an infinitely long line charge having linear charge density (λ) is 3.6×10^3 V/m. Find the value of λ .
13. Two point charges $Q_1 = 400 \mu\text{C}$ and $Q_2 = 100 \mu\text{C}$ are kept fixed, 60 cm apart in vacuum. Find intensity of the electric field at midpoint of the line joining Q_1 and Q_2 .
14. A large hollow metallic sphere has a positive charge of $35.4 \mu\text{C}$ at its centre. Find how much electric flux emanates from the sphere.
15. A point charge of 5×10^{-6} C experiences a force of 2×10^{-3} N when kept in a uniform electric field of intensity E. Find E.
16. A point charge of $2.0 \mu\text{C}$ is at the centre of a cubic Gaussian surface 9.0 cm on edge. What is the net electric flux through the surface? What if the charge is not at the centre?
17. Four point charges are placed at the corners of a square of side 2 cm, as shown. Find the magnitude and direction of the electric field at the centre O of the square, if $q = 0.02 \text{ mC}$.



18. Charges of $+1.2 \times 10^{-8}$ C and -1.6×10^{-8} C are placed at two points A and B respectively, distant 5 cm from each other. Compute the electric field at a point C distant 3 cm from A and 4 cm from B.
19. A charged oil drop weighing 1.6×10^{-15} N is found to remain suspended in a uniform electric field of intensity 2×10^3 N/C. Find the charge on the drop.
20. Two infinitely large plane thin parallel sheets having surface charge densities σ_1 and σ_2 ($\sigma_1 > \sigma_2$) are shown in the figure. Write the magnitudes and directions of net fields in the regions marked I, II and III.



PRACTICE PROBLEMS

1. A polythene piece rubbed with wool is found to have charge of $-3 \times 10^{-7} \text{ C}$.
(i) Estimate the number of electrons transferred (from which to which)?
(ii) Is there a transfer of mass from wool to polythene?
 2. Two identical metal spheres having equal and similar charges repel each other with a force of 1800N when they are placed 10cm apart in a medium of dielectric constant 5. Determine the charge on each sphere.
 3. A charge of $17.7 \times 10^{-4} \text{ C}$ is distributed uniformly over a large sheet of area 200 m^2 . Calculate the electric field intensity at a distance of 20 cm from it.
 4. A particle of mass $5 \times 10^{-6} \text{ g}$ is kept over a large horizontal sheet of charge density $4 \times 10^{-6} \text{ C/m}^2$. What charge should be given to this particle so that if released it does not fall down? How many electrons should be removed to give this charge?
 5. Two point charges of $+16 \mu\text{C}$ and $-9 \mu\text{C}$ are placed 8 cm apart in air. Determine the position of the point at which the resultant electric field is zero.
 6. Calculate the electric field strength required just to support a water drop of mass 10^{-7} kg and having a charge $1.6 \times 10^{-19} \text{ C}$.
 7. Two point charges of $2 \mu\text{C}$ and $8 \mu\text{C}$ are placed 12 cm apart. Find the position of the point where the electric field intensity will be zero.
 8. Two charges $+20 \mu\text{C}$ and $-20 \mu\text{C}$ are placed 1 cm apart. Calculate the electric field at a point on the axial line at a distance of 20 cm from the centre of the dipole.
 9. Two pith balls each of mass 30 mg are suspended by two threads from a support. When charged identically, the two pith balls move apart by 0.08 m, the threads making angle 60° with each other. Calculate the charge on each pith ball.
 10. Given a uniform electric field $E = 4 \times 10^3 \text{ i N/C}$. Find the flux of this field through a square of 5 cm on a side whose plane is parallel to the Y-Z plane. What would be the flux through the same square if the plane makes a 30° angle with the X-axis?
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